Will Vance

Broomfield, CO — wjvance213@gmail.com — wjvance.com — github.com/wjvance213

Summary

Computer Science undergraduate at CU Boulder building real-time software across computer vision, audio plugins, and web development in Python, C++, and JavaScript. Seeking a Summer 2027 software engineering internship.

Education

University of Colorado Boulder — GPA: 3.14 *Expected Graduation: Spring 2028*
Bachelor of Science in Computer Science

Skills

Programming Languages: C, C++, Python, JavaScript, SQL, HTML/CSS, Bash
Frameworks/Libraries: OpenCV, MediaPipe, JUCE, React, Vite, Blender Python API (bpy)
Tools: Git, Docker, PostgreSQL, Linux, VSCode, Visual Studio, Arduino
AI Tools: Agentic coding workflows with Claude Code

Projects

Body Tracker – Motion Capture (Python, OpenCV, MediaPipe) — GitHub *Fall 2026 – Present*

- Built a real-time motion-capture pipeline in Python that tracks up to 46 hand and body landmarks from a webcam with MediaPipe and streams them to a Blender add-on that drives a 47-bone rig live
- Implemented a One Euro filter to smooth noisy landmark data, cutting jitter ∼35% versus a moving average while reducing lag from 163 ms to 31 ms
- Designed a versioned JSON-over-UDP protocol with sequence numbers and timestamps (≤2.6 KB/frame) plus a listener that detects lost or out-of-order packets, sustaining ∼28 packets/s with zero loss locally
- Developed a Blender extension that polls a non-blocking socket from Blender's main thread (bpy is not thread-safe) and generates an armature with bone constraints programmatically

SquillysDelay – Tape-Echo Audio Plugin (C++, JUCE) — GitHub *Fall 2026 – Present*

- Built a tape-echo delay plugin (VST3 + standalone) in C++ with JUCE 8 for FL Studio, around a per-channel circular-buffer delay line sized up front so the real-time audio thread never allocates, locks, or does I/O
- Made delay-time changes click-free with fractional reads, linear interpolation, and a 300 ms glide, cutting the largest sample-to-sample jump in testing from 0.97 to 0.019; smoothed all other parameters over 20 ms
- Added a hand-written sine LFO that modulates delay time (0.05–10 Hz, up to ±10 ms) for chorus, flanger, and tape-warble sounds, with 0–180° stereo spread
- Synced delay time to host tempo through the plugin playhead API with 14 note divisions (straight, triplet, dotted) and a safe fallback when the host reports no tempo
- Wrote an offline test harness that drives the audio callback directly, checks every echo's position and amplitude, tracks the modulated read position to within 0.01 samples, and confirms its click checks fail when smoothing is removed

Personal Website – wjvance.com (React, Vite, JavaScript, CSS, Cloudflare) — GitHub *Fall 2026*

- Built a responsive single-page portfolio in React 19 and Vite, rebuilt from an earlier plain-HTML class site, with continuous deployment through Cloudflare on every push to main
- Wrote a reusable scroll-reveal component using the IntersectionObserver API with staggered CSS transitions and reduced-motion support
- Kept all site content in a single data module and the theme in CSS custom-property design tokens, so copy, links, and colors update without touching component code
- Kept the bundle to ∼65 KB gzipped JS by using dependency-free inline SVG icons and replacing a demo GIF with a ∼425 KB looping MP4

Work Experience

Rocky Mountain Chocolate Factory *2022–2024*
Sales Associate

- Handled POS transactions, cash, and inventory in a fast-paced retail and food-service environment

Action Behavior Centers *June 2024–Oct 2024*
Registered Behavioral Therapist

- Delivered structured behavioral interventions for children with Autism Spectrum Disorder in a clinical setting
- Documented session data in HIPAA-compliant software and coordinated care plans with board-certified behavior analysts